MILITARY SPECIFICATION

TRANSDUCER, PRESSURE

This specification is approved for use by all
Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification establishes the requirements for the procurement of a Pressure Transducer, NOVORD Outline Drawing 1803636.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of incitation for bids or request for proposal form a part of this specification to the extent specified herein.

* SPECIFICATIONS

Military

MIL-P-116	Rwservation- Packaging, Methods of
MIL-E-82590	Enviromental Requirements and Tests (For Torpedo Mk 37 Components and Assemblies)

* STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base
MIL-STD-810	Environmental Test Methods

FSC 1355

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DRAWINGS

Naval Ordnance Systems Command

LD 620031	shipping (Packing) Condition
Dwg 1803636	Transducer, Pressure

* PUBLICATIONS

Naval Ordnance Systems Command

os 6341	Miscellaneous General Ordnance Design Requirements

(Copies of specifications, standards, drawings, and publications requested by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publiacations. The following documents farm a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards Publication

Handbook H28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTSS

3.1 Description. The transducer is a pressure-sensitive device which produces a voltage output proportional to the external pressure applied to the unit. The transducer incorporates a pressure-sensitive tube, to the outside of which are bonded resistive strain wire grids in abridge network. Variations in pressure cause the grids to expand and stretch, changing the resistance. Therefore, each change in pressure will cause a change in voltage output.

3.2 General requirements

* 3.2.1 Data required. No data is required by this specification (other than reports accompanying samples submitted for preproduction and periodic

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production testing), or by applicable documents referenced in Section 2," unless specified in the contract or order (see 6.2).

3.2.2 Preproduction sample. Unless otherwise specified in the contract or order, preproduction samples of the transducer shall be manufacured using the methods and procedures proposed for the production lot. These units shall be tested as specified in Section 4 herein and are for the purpose of determining, prior to starting production, that the contractor's production methods are capable of producing transducers that comply with the technical requirements of the contract. See 4.3.1.

3.2.3 Materials. Unless otherwise specified on the requisition, order, or contract, all materials and purchased parts entering into the final assembly of the transducer shall conform strictly to the detail requirements indicated on Drawing 1803636.

3.2.4 Finishes, coatings, construction} and dimensions. Finishes, coatings, construction, and dimensions for the transducer shall be strictly In accordance with Drawing 1803636.

3.2.5 Moisture, fungus, and corrosion. The transducer shall be rendered moisture, fungus, and corrosion resistant in accordance with OS 6341.

3.2.6 Marking. Marking of assemblies, subassemblies, and parts shall be in accordance with the applicable drawings listed in Drawing 1803636.

3.2.7 Threads. Unless otherwise specified, all threads shall be in accordance with National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

* 3.2.8 Conflicting requirements. Conflicting requirements arising between this specification and any specifications, publications. or drawings listed herein shallbe referred in writing to the procuring agency or appointed agent for interpretation, clarification, and resolution (or correction).

* 3.3 Definitions. Definitions applicable to the environmental requirements of this specification are set forth in MIL- E-825900

3.3.1 Full-scale output. Full-scale is defined as the output voltage obtained upon application of 500 psig and 8 volts to the pressure transducer.

3.3.2 Linearity. Linearity is the maximum deviation of the calibration curve from a straight line drawn between the no-pressure and full-scale pressure outputs, compressed as a percentage of the full-scale outpur and measured on Increasing pressure only.

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3.3.3 Hysteresis. Hysteresis is the maximum difference between pressure cell output readings for the same applied pressure, one reading obtained by increasing the pressure from zero and the other by decreasing the pressure from full scale.

3.4 Performance requirements and product characteristics. The transducer shall meet the following performance requirements and product C~-acteristics:

3.4.1 Operating range. The transducer shall operate satisfactorily when subjected to hydraulic pressures ranging from O psig to 500 psig.

3.4.1.1 The transducer shall require no recalibration when subjected to at least 600 psig.

3.4.2 Operating voltage.e. The transducer shall operate satisfactorily when energized by an input voltage of 8 ± .4 volts, 400 ± 12 Hertz (Hz).

3.4.3 Zero drift. Zero drift, due to the application of 8 ± .4 volts, 400 ± 12 HZ to terminals 3 and 4, shall not exceed 0.25 percent of full-scale output at the 15-second period after application of voltage. After 15 seconds and for a 5 minute period thereafter, the zero drift shall not exceed 0.1 percent of full-scale output and shall continue to drop thereafter until thermal equilibrium is reached, when tested as specified in 4.5.4.2.

3.4.4 Ourput voltage. The output voltage of the pressure cell shall be 1.95 microvolt plus 15 percent minus 10 percent per volt input per psig.

3.4.5 Zero error. The zero error shall be less than 5 percent of full-scale output.

3.4.6 Maximum nonlimearity. For increasing pressure, the maximum nonlinearity shall be less than plus or minus 0.25 percent of full-scal output.

3.4.7 Hysteresis. The difference in outputs obtained by increasing the pressure from 0 to 500 psig and decreasing the pressure from 500 to O psig shall be within 0.25 percent of full-scale output.

3.4.8 Zero return. The zero return after a one minute pressure cycle from O to 500 psig and return shall be within 0.05 percent of full-scale output. Fifteen seconds after return to zero, the zero return shall decrease to 0.01 percent of full-scale output.

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3.4.9 Resolution. Resolution shall be 0.01 percent of full-scale output.

.3.4.10 Repeatability. Repeatability shall be within 0.1 percent of full-scale output. Use the average of corresponding data points in a comparison of two successive pressure cycles. Test in accordance with 4.5.4.7.

3.4.11 Resistance. With no pressure applied, resistance shall be as follows :

Input (terminals 3 and k):	247 ohms plus or minus 2 percent
Output (terminals 1 and 2):	234 ohms plus or minus 2 percent
Between terminals 3 and 1:	176 ohms plus or minus 2 percent
Between terminals 3 and 2:	176 ohms plus or minus 2 percent
Between terminals 4 and 1:	185 ohms plus or minus 2 percent
Between terminals 4 and 2:	185 ohms plus or minus 2 percent

3.4.12 Insulation resistance. With 50 volts dc applied, insulation resistance shall be 300 megohms minimum when tested as specified in 4.5.4.12.

* 3.5 Environmental requirements. The transducer shall meet all requirements of this specification during or after the tests defined or specified in 4.4,

3.5.1 Vibration! The transducer shall meet the following vibration requirements:

* 3.5.1.1 Vibration, nonoperating. The transducer shall. show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.1.1.

* 3.5.1.2 Vibration, operating. The transducer shall show no damage and shall meet all the performance requirements and product characteristics of this specification during and after the test specified in 4.4.1.2.

*

* 3.5.2 Temperature and humidity

* 3.5.2.1 High temperature and humidity, nonoperating. The transducer shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.2.1

* **3.5.2.2 High temperature, operating. The transducer shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.4.2.2.**

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* 3.5.2.3 Low temperature, nonoperating. The transducer shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.2.3.

* 3.5.2.4 Low temperature, operating. The transducer shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.4.2.4.

3.5.2.5 Thermal effect on zero shift. With input voltage at 8 volts, zero shift for thermal equilibrium within the cell shall be less than 0.01 percent of fullscale output per degree Fahrenheit (F) throughout a temperature range of plus 28 degrees to plus 130 degrees F,

3.5.2.6 Thermal effect on sensitivity shift. Sensitivity shift for thermal equilibrium within the cell shall be less than 0.01 percent of full-scale output per degree F throughout a temperature range of plus 28 degrees to plus 130 degrees F.

3.5.2.7 Thermal shock. Wetting of the pressure sensing element with water at a differential temperature of 40 degrees F followed by immersion of the transducer shall not change the O psig pressure output by more than 4 percent of full-scale output within a period of one minute after immersion. After one minute of immersion, the change shall not exceed 0.5 percent of full scale output.

* 3.5.3 Fungus. The transducer shall meet all performance requirements of this specification after being tested in accordance with 4.4.3.

3.5.4 Pressure-voltage sensitivity. The maximum time for the output voltage to reach 63 percent of the final output for any step pressure input, within the range of the transducer, shall be 2.4 milliseconds.

3.5.5 Zero output stability. The transducer output at O psig, with zero unbalance corrected for minimum, shall not change by more than 0.05 percent of full-scale output when the source voltage is varied over the range of 7.76 to 8.24 volts.

3.5.6 Phase relationships.

3.5.6.1 At O psig, the portion of the output voltage that is 90 degrees to the supply voltage shall not exceed plus or minus 1.5 microvolt output per volt input at 400 Hz.

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3.5.6.2 The phase change in output voltage for a given pressure stimulus shall not exceed an angle of plus or minus 0.1 degree with respect to the supply voltage.

3.5.6.3 When pressure is applied, output terminal 1 shall have the same polarity as input terminal 4.

3.6 Cleanup. Prior to and after final assembly, all parts, components, and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign rotter. Burrs and sharp edges as well as rosin flash shall be removed.

3.7 Workmanship. The transducer, including all parts and accessories, shall be constructed and finished in a manner to ensure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, painting, riveting, machine screw assemblage, welding, brazing, and freedom of parts from burrs and sharp edges. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where-such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections. Inspection of the pressure tansducer shall be classified as follows:

(a) Preproduction inspection (paragraph 4.4)

(b) Quality conformance inspection (paragraph 4.5)

(c) Periodic production inspection (paragraph 4.4)

4.3 Sampling

4.3.1 Reproduction. A preproduction sample of three pressure transducers, manufactured in accordance with 3.2.2, shall be subjected to the preproduction tests detailed in 4.4 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the transducer by the supplier, prior to the approval of the preproduction sample, shall "be at the supplier's risk.

4.3.2 Lot size. Unless otherwise specified in the contract or order, each shipment of units of product shall be a separate "inspection lot" and may differ from the quantity specified in the total order or contract. In any event, all units of a single "inspection lot" shall be made to the same design, with the same materials and manufacturing techniques, and shall be part of one continuous production run.

* 4.3.3 Periodic production samples. From each succeeding lot of 3200or less hace passed the quality conformance inspection of 4.5, a random sample of five transudcers will be selected for submission to an activity designated by the procuring activity to determine compliance of the samples with the requirements of the contract, specification, and drawings. The transducers shall be considered satisfactory if they meet all specification and pertinent drawings requirements and withstand the tests specified in 4.4 and 4.5. Failure of the periodic production sample to comply with these requirements will result in the rejection of the lot, as determined by the procuring activity.

* 4.4 Preproduction and periodic production inspection. The preproduction and periodic production samples, after satisfactorily passing the quality conformance inspections detailed in 4.5, shall be subjected to the following tests and examinations. Certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the procuring activity.

* 4.4.1 Vibration

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4.4.1.1 Vibration, nonoperating. The transducer s-hall be subjected to the "Vibration, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.5.1.1.

* 4.4.1.2 Vibration, operating. The transducer shall be subjected to the "vibration, operating" test of MIL-E-82590 and during and thereafter shall meet the requirements of 3.5.1.2.

4.4.2 Temperature and humidity

* 4.4.2.1 High temperature and humidity, nonoperating. The transducer shall be subjected to the "High temperature and humidity, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.5.2.1.

* 4.4.2.2 High temperaturem operating. The transducer shall be subjected to the "High temperature, operating" test of MIL-E-82590 and, while at the test temperature, shall meet the requirements of 3.5.2.2.

* 4.4.2.3 Low temperature, nonoperating. The transducer shall be subjected to the Low temperature, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.5.2.3.

* 4.4.2.4 Low temperature, operating. The transducer shall be subjected to the "Low temperature, operating" test of MIL-E-82590 and, while at the test temperature, shall meet the requirements of 3.5.2.4.

* 4.4.2.5 Thermal effect on zero and sensitivity shift. With 8 volts, 400 Hz applied, the transducer shall be placed within a test chamber and subjected to the following temperatures and pressures for a period of one hour at each temperature setting:

0	+ 28°F
500	+ 28°F
0	+130°F
500	+130°F

The transducer shall meet the requirements of 3.5.2.5 and 3.5.2.6.

4.4.2.6 Thermal shock. With no pressure, 8 volts, 400 Hz applied to the transducer, measure and record the output voltage. Prior to immersion, ensure that the pressure tap of transducer is open and that the cavity is filled with water at a differential temperature of 40 degrees F. Completely submerge the transducer in water at a differential temperature of 40 degrees F from that of the transducer, for a one minute period. Measure and record the output voltage at 10, 20, 30, and 45 seconds, and
* at one, two, three and five minutes after immersion. The transducer shall meet the requirements of 3.5.2.7.

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4.4.3 Fungus. The transducer shall be tested for resistance to fungus in accordance with method 508, procedure I of MIL-STD-810 and, thereafter, shall meet the requirements of 3.5.3.

4.4.4 Pressure-Voltage sensitivity. Mount the transducer in a shock tube and energize input terminals with a stable power supply. Output shall be monitored on an oscilloscope with a suitable camera attachment. A step pressure input (shock) shall be applied to the transducer and the output voltage rise shall be photographed. The output shall meet the requirements of 3.5.4.

4.4.5 Zero output stability. Connect two 3000 ohm wire-wound resistors in series across output terminals 1 and 2 with the midpoint grounded. Connect a 25,000 ohm potentiometer across output terminals 1 and 2, with a 200,000 ohm resistor connected between the slider and input terminal 3. Apply 8 volts, 400 Hz or 8 volts dc to the input of the transducer and adjust the slider for minimum output. With 0 psig applied to the transducer, vary the source voltage over the range of 7.76 to 8.24 volts. The transducer shall meet the requirement of 3.5.5.

4.4,6 Using the test circuit of Figure 1, simulate the pressure cell with four 120 ohm, nonreactive film resistors and measure the quadrature of the test circuit as indicated on the oscilloscope (simulates transducer at 0 psig). Shunt one arm of the simulating network with a 60,000 ohm, nonreactive film resistor and measure the quadrature of the test circuit (simulates transducer at 500 psig). Record these two readings, Remove the simulating resistors and connect the pressure transducer. With 0 psig applied, measure the quadrature of the circuit. Apply 500 psig to the transducer and measure the quadrature voltage. Record these two readings. Determine the quadrature of the transducer at 0 psig by subtracting the value obtained at 0 psig with the simulating network from that obtained with the transducer. The transducer shall meet the requirement of 3.5.6.1. Record this value. The phase angle of the output voltage with pressure applied may then be determined by the following formula:

$$\text{Sine of Phase Angle} = \frac{(RV_{500} - RV_0) - (RV_{T500} - RV_{TO})}{\text{Output Voltage in Microvolts at 500 psig}}$$

*RV_{500} - Reactive Voltage of the Cell at 500 psig

RV_0 - Reactive Voltage of the Cell at 0 psig

RV_{T500} - Reactive Voltage of the Test Circuit, equivalent to RV_{500}

RV_{TO} - Reactive Voltage of the Test Circuit at 0 psig

*All voltagegs expressed in microvolts.

The transducer shall meet the requirement of 3.5.6.2.

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4.5 Quality conformance inspection. Each pressure transducer delivered under contract shall be subjected to the following tests and examinations. Failure to meet any requirement specified herein shall be considered cause for rejection.

4.5.1 Test conditions. Unless otherwise specified, the transducer shall be subjected to quality conformance tests under the following conditions:

4.5.1.1 Temperature. Room ambient 65 degrees F to 95 degrees F (18 degrees c t - e s C) .

4.5.102 Altitude. Normal ground.

4.5.1.3 Vibration. None

4.5.1.4 Humidity. Room ambient to 95 percent relative (maximum).

4.5.2 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities and shall provide personnel for performing all quality conformance tests. The test equipment shall be adequate in quantity and, when definite requirements are not specified, shall be of sufficient accuracy and quality to permit performance of the required tests.

4.5.3 Test equipment. The following items of test equipment are required to perform the quality conformance tests set forth in this specification:

4.5.3.1 Calibration indicator, Baldwin Lima-Hamilton Model 615 (modifed to provide an 8± .4 volt, 400 ± 12 Hz output signal) or equivalent.

NOTE

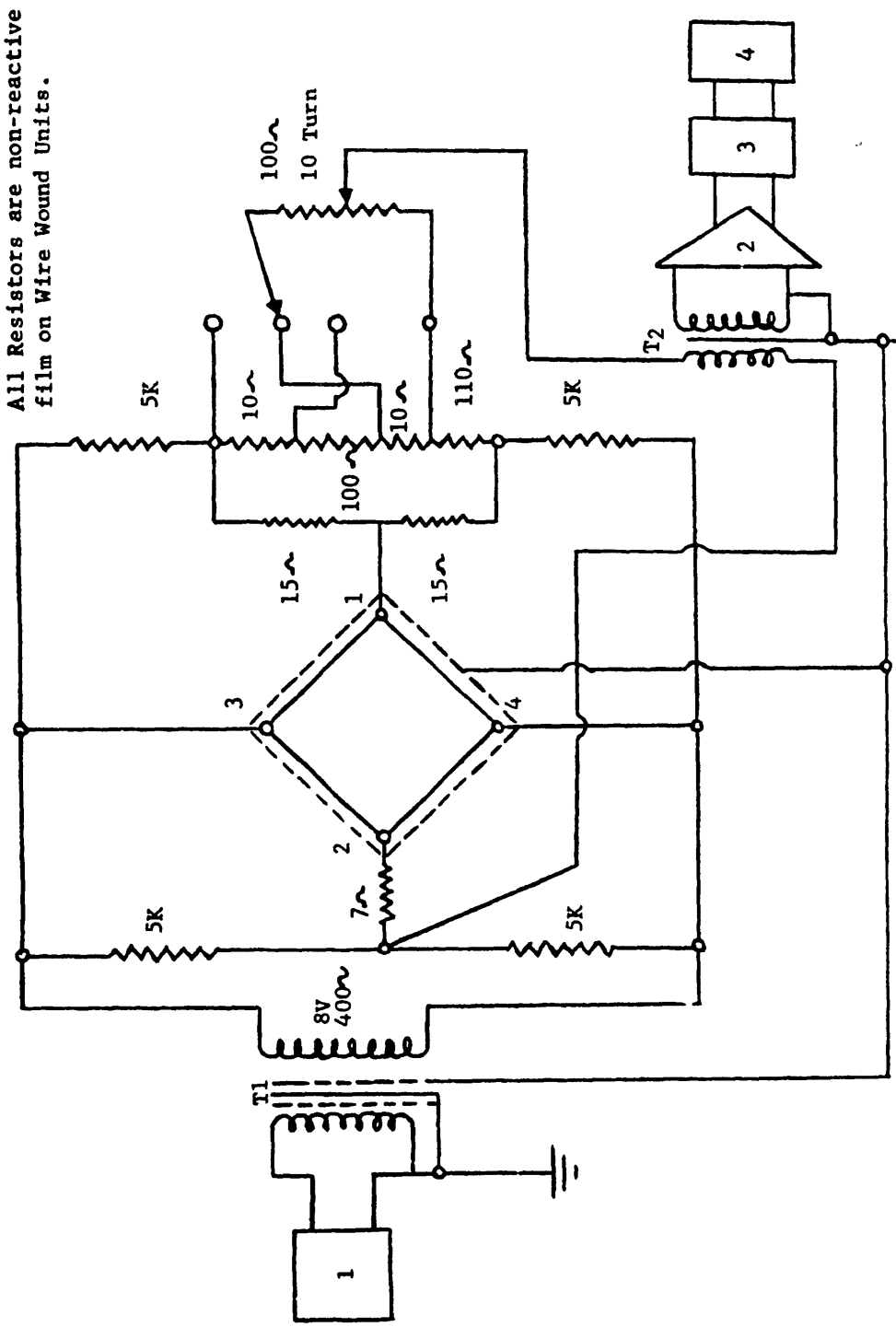
DC excitation may be used for all tests except the phase shift tests of 3.5.8.

4.5.3.2 Resistance bridge, Leeds and Northrup Type 4735, or equivalent. Accuracy: 0.05 percent.

4.5.3.3 Megohmmeter, capable of measuring insulation resistance at 50 volts dc over a range of O to 500 megohms.

4.5.3.4 Primary Pressure Standard, Mansfield and Green, Model RQ-100, or equivalent.

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1 = Audio Signal Generator H.P. 205 AG or Equivalent.
2 = Kintel 112A or Equivalent.
3 = Filter Krohn-Hite 330A or Equivalent.
4 = H.P. 130B or Equivalent.

T_1 - UTC Type HIT-1

T_2 - Minneapolis-Honeywell # 355567-1

Figure 1

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4.5.3.5 Glass cylinder capable of supplying a hydrostatic pressure head of at least 6 inches.

4.5.4 Test procedure

4.5.4.1 Connect the transducer to the calibration indicator (4.5.3.1).

4.5.4.2 zero drift. With no pressure applied, energize the transducer and record the output at 5, 10, and 15 second intervals and at one minute intervals thereafter for 5 minutes. The transducer shall meet the requirement of 3.4.3.

4.5.4.3 Zero error. At no time during zero pressure application shall the transducer exceed the zero error requirement of 3.4.5.

4.5.4.4 Output voltage

NOTE

The test results for output voltage, linearity, hysteresis, and repeatability may be obtained by conducting three successive pressure cycles in accordance with the procedures and tests outlined below.

4.5.4.4.1 Apply 500 psig to the transducer and record the output. The output voltage shall meet the requirement of 3.4.4.

4.5.4.4.2 During the test of 4.5.4..4.1, the canter shall give increasing readings to verify the requirement of 3.5.8.3.

4.5.4.5 Maximum nonlinearity. IN 100 psig increments, increase the pressure from O to 500 psig. Record the output at each increment. The transducer shall meet the linearity requirement of 3.4.6.

4.5.4.6 Hysteresis. In 100 psig increments, decrease the pressure from 500 to O psig. Record the output. Using these values and the values obtained in 4.5.4.5, determine the hysterisis. The transducer shall. meet the requirement of 3.4.7.

4.5.4.7 Repeatability. Load the pressure cell to 250 and 500 psig and return to 250 and O psig. Record the output at each increment. Perform this test a total of three times. The transducer shall meet the requirement of 3.4.10.

4.5.4.8 Zero return. With O psig applied to the transducer, record the output voltage. Cycle the transducer from 0 to 500 psig and return

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to O psig within one minute. Record the O psig output immediately on return to O psig and 15 seconds later. The transducer shall meet the requirements of 3.4.8.

4.5*4.9 Operating range. Cycle the pressure cell from 0 to 500 psig and record the output at these extremes. Load the pressure cell to 600 psig and then return to zero. Repeat the O to 500 psig cycle and record the output at the extremes. The difference between the readings-obtained before and after the 600 psig cycle shall be not greater than 0.1 percent of full scale to verify the requirement of 3.4.1.1

4.5.4.10 Resolution. Disconnect the primary pressure standard from the transducer and couple the glass cylinder to the transducer. Fill the cylinder with enough water to move the cell off zero. Record the output. Increase water head by 1-3/8 inches (equivalent to .05 psi) and record the output. The transducer shall meet the requirement of 3.4.9.

4.5.4.11 Disconnect the transducer from the glass cylinder and the calibration indicator.

4.5.4.12 Insulation resistance. Connect transducer terminals 1, 2, 3, and 4 together, using the megohmmeter (4.5.3.3), and with 50 volts dc applied, measure resistance between the terminals and case and between terminals and ground. The transducer shall meet the requirement of 3.4.12.

4.5.4.13 Resistance. Disconnect transducer terminals 1, 2, 3, and 4 from each other and, with no pressure applied to the transducer, measure resistance between terminals, using resistance bridge (4.5.3.2). The resistance shall be as specified in 3.4.11.

4.5.4.14 Visual and dimensional examination. Each transducer shall be examined to determine conformance with the requirements for workmen0 ship and dimensions.

4.5.4.15 Packaging, paeking, and marking. Packaging, packing, and marking shall comply with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

* 5.1 reservation and packaging. The levels of preservation and packaging shall be Level A or C as specified (see 6.2).

* 5.1.1 Level A

* 5.1.1.1 unit packaging. The pressure transducer shall be packaged in accordance with Method 1A-5 of MIL-P-116 and LD 620031.

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5.1.2 Level B. Not applicable.

5.1.3 Level C

5.1.3.1 Unit packaging. The pressure transducer shall be packaged in accordance with Method III of MIL-P-336.

5.2 Packing. Packing shall be Level A or Level C as specified (see 6.2).

5.2.1 Number of packages per pack. The number of units packages per pack, for Level C, shall be as specified in the contract or order.

* 5.2.2 Level A

* 5.2.2.1 Unit pack. The pressure tansducer shall be packed in accordance with LD 20031. and may be shipped individually in small lots.

* 5.2.2.2 Palletizing. When specified, Level A packs shall be palletized in accordance with MIL-STD-147, Load Type XIII (see 6.2).

5.2.3 Level B. Not applicable.

5.2.1 Level C. The pressure transducer shall be packed to afford protection against damage during direct shipment from the supply source to the first receiviug activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

* 5.3 Marking. Marking shall be Level A or C as specified (See 6.2)

* 5.3.1 Normal markings, Level A. Unit packs shall be marked in accordance with the requirements of MIL-STD-129 and LD 620031.

* 5.3.2 Normal markings, Level C. Unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

* 5.3.3 Serial numbers. Unit packages and shipping containers shall be marked with the serial. numbers of the units contained within.

6. NOTES

6.1 Intended use. The transducer is intended for use in Torpedo Mk 37 Mods 0, 2, and 3.

* 6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement Requirements

(a) Title, number, and date of this specification.

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(b) Any exceptions to this specification.

(c) Required level of preservation and packaging (see 5.1).

(d) Required level of packing including palletizing (see 5.2).

(e) Required level of marking (see 5.3)

(f) Assigned activity for preproduction and periodic production inspections.

(g) Data required other than that specified in 3.2.1. (see 4.4)

(h) That additional samples required because of failure of any unit(s) to pass the prescribed periodic production inspections shall be provided and inspected at the expense of the supplier.

6.2.2 Contract Data Requirements. Any data required for delivery in connection with this document shall be specified on a DD Form 1423 incorporated into the contract. Such data will be delivered as identified on completed (numbered) DIDs (Data Item Descriptions/DD Form 1664) which will be documented in the applicable ADL (Authorized Data List).

6.3 Preparation for delivery criteria. Criteria for use of the proper level of preservation, packaging, and packing shall be as follows:

Level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

Level C. This level shall be used only when it is definitely known that the packaged items are to be shipped to domestic installations for immediate use at the first receiving activity.

6.4 Marginal identification. The margins of this specification are marked with an asterisk * to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Navy - OS

Preparing Activity:
Navy - OS
Project No. 1355-N168

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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DEPARTMENT OF THE NAVY

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NAVY DEPARTMENT

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Commander, Naval Sea Systems Command
Engineering Support Division
Attn: SEA-981
Department of the Navy
Washington, DC 20360

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